ALICE L WALTON SCHOOL OF MEDICINE

Introduction to Virology

Last updated December 8, 2025 17 min read
CC BY-SA 4.0

Adapted by: Ray Brown, Lori Garman
Editor: Jordan Finn

Adapted from: Virology: Foundations and Frameworks

Listen to this Brick
0:00 / 28:24 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe how common structures and genetic information are used to categorizes viruses into families.
2. Explain the key steps of a viral infection.
3. Compare and contrast the replication cycle of DNA, (–)ssRNA, and (+)ssRNA viruses.
4. Compare and contrast the common lab assays used to identify viral infections and their limitations.

CASE CONNECTION

A 27-year-old male presents with a four-day history of cough, mild fever, fatigue, and anosmia (loss of smell). Many of his coworkers have been absent due to similar symptoms. While COVID-19 is suspected, confirmation is necessary as various respiratory pathogens have recently been prevalent in the community.

Based on this patient's symptoms and the time they have been ill, which test would be the most appropriate to confirm the diagnosis? Which test would be inconclusive at this stage of illness?

GO TO CONCLUSION ↓

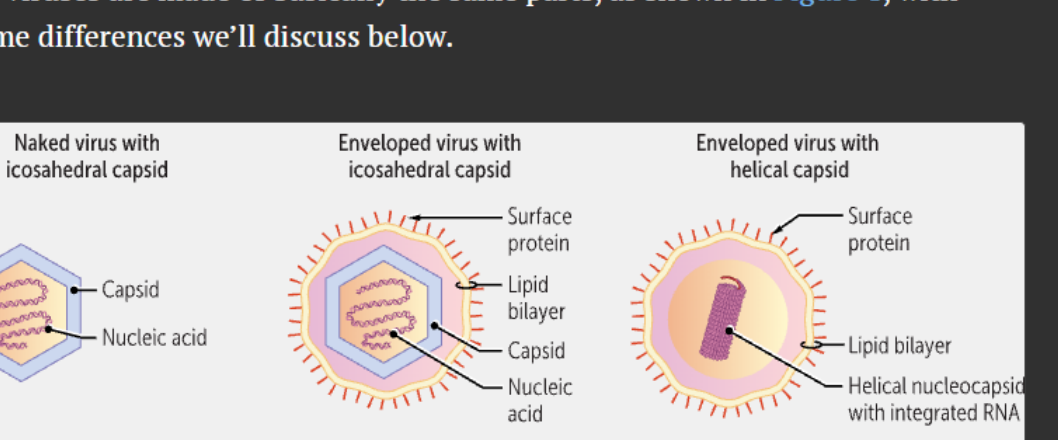
What Is a Virus? (LO1)

For centuries, the prevailing medical theory was that many diseases were caused by breathing bad air. Even with the discovery of bacteria, what we now call the germ theory did not catch on until the late 19th century. Yet germ theory could not explain all diseases, because in some infections, no organism could be found. We later learned that some diseases were caused by particles too small for microscopes to show; we now call those particles viruses.

Viruses are a nearly ubiquitous form of early proto-life—small particles of genetic code, lipids, and proteins that straddle the line of living and nonliving. Viruses infect all livi[illegible] to the largest animals and plants. They are a major driver of the evolution of all life. Because they can't reproduce on their own, they are not considered truly alive. But being nonliving doesn't stop them from being one of the greatest problems we face in modern medicine.

Virus Types

All viruses are made of basically the same parts, as shown in Figure 1, with some differences we'll discuss below.


Naked virus with icosahedral capsid
Capsid
Nucleic acid
Enveloped virus with icosahedral capsid
Surface protein
Lipid bilayer
Capsid
Nucleic acid
Enveloped virus with helical capsid
Surface protein
Lipid bilayer
Helical nucleocapsid with integrated RNA

QUIZ

Figure 1

All viruses have some sort of genetic code, either DNA or RNA, and a protein coat called a capsid. Capsids are usually icosahedral or helical in shape. Some viruses are also covered by a lipid bilayer called an envelope, usually a remnant of cell membrane from the host cell. All non-enveloped viruses are icosahedral and stable in the env[illegible]

Viruses are mainly classified by the type of nucleic acid they contain—single stranded or double stranded RNA or DNA. The genome is associated with what enzymes they carry with them, like the enzyme reverse transcriptase (RT), which produces DNA from RNA. Viruses are also classified by whether or not they are enveloped and the shape of their capsid.

How are viruses categorized?

Virus Classification System

Very specific viral classification systems exist, with the Baltimore system being the most widely used. The most clinically relevant classification is into three classes: DNA viruses, RNA viruses, and retroviruses.

DNA viruses, as the name suggests, use DNA as their genome. All use double stranded (dsDNA) exc[illegible]tranded (ssDNA). Their genome must be transcribed into messenger RNA (mRNA) prior to viral replication.

Almost all RNA viruses use single-stranded (ssRNA) RNA for their genome; reoviruses use dsRNA. Single-stranded sense ssRNA, or (+)ssRNA, is equivalent to mRNA and can be translated directly to protein without forming an intermediate molecule. Antisense ssRNA, or (–)ssRNA, is first transcribed to (+)ssRNA, which is then translated to protein.

Retroviruses are (+)ssRNA viruses (ssRNA-RT); the most infamous of these is human immunodeficiency virus (HIV). Retroviruses carry the enzyme reverse transcriptase (RT), which transcribes their RNA into DNA (hence "reverse" transcriptase). These virally produced DNA transcripts are then read by the infected cell as if it were host DNA.

Which RNA molecule can be translated into a protein directly, (–)ssRNA or (+)ssRNA?

How Do Viral Infections Occur? (LO2)

For all these viruses we've discussed, the end goal is the same: get into a cell, take over its machinery, and make the cell into a virus-production factory. However, the way these viruses accomplish this goal is highly variable.

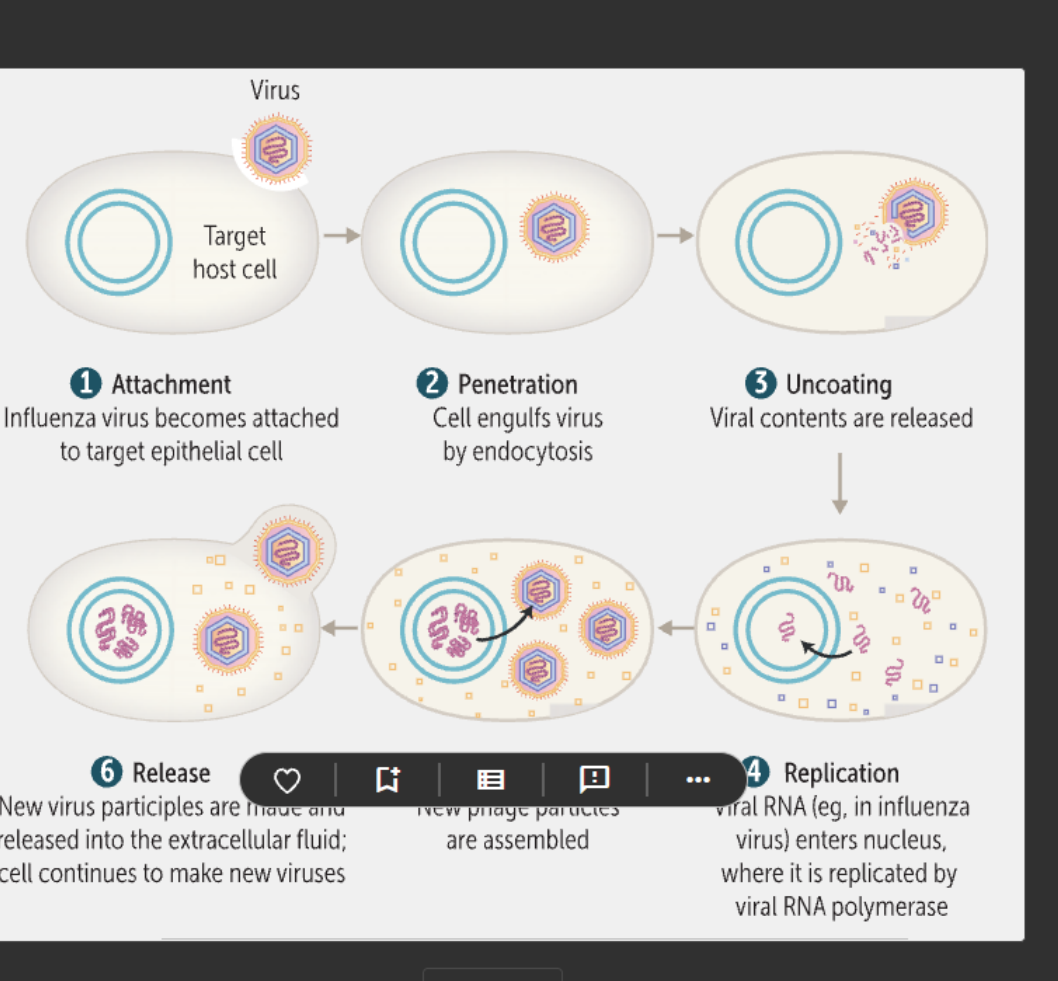
Steps of Viral Infection

As a general rule, viruses follow a step-by-step procedure during an infection:

- **Attachment:** virions (viral particles) attach to the cell surface, usually using a surface marker or receptor.
- **Penetration:** The virus enters the cell, usually either via endocytosis or fusion of the viral envelope with the cell membrane.
- **Uncoating:** The virus loses its protein coat, leaving the nucleic acids and enzymes it may be carrying.
- **Replication:** Viral nucleic acids, enzymes, and proteins are replicated using host machinery. Most DNA viruses replicate in the nucleus. Most RNA viruses replicate in the cytoplasm.
- **Assembly:** New viral nucleic acids, enzymes, and proteins assemble to form new virions.
- **Release:** The cell releases mature virions (either by exocytosis/budding or host cell lysis and death).

What are the six steps of a viral infection of a cell?

If this seems almost mechanical, it's because it truly is. Think of viruses as little machines with a very specific task—replication (Figure 2).


Virus
Target host cell
1 Attachment
Influenza virus becomes attached to target epithelial cell
2 Penetration
Cell engulfs virus by endocytosis
3 Uncoating
Viral contents are released
4 Replication
Viral RNA (eg, in influenza virus) enters nucleus, where it is replicated by viral RNA polymerase
New phage particles are assembled
6 Release
New virus participles are made and released into the extracellular fluid; cell continues to make new viruses

QUIZ

Figure 2

How Do Viruses Know Which Cells to Infect?

Why do some viruses affect some organs but leave others alone? For example, why does the influenza virus primarily affect the respiratory system while HIV targets the immune system? The answer is receptors and the viruses' affinity to specific receptors on host cells. For some viruses, the cell receptor they target is highly specific for a particular cell type or tissue. These target receptors are a major part of what we call viral tropism (basically, the selection of tissues a virus can target).

HIV, for example, targets CD4 T lymphocytes, monocytes/macrophages, and dendritic cells. HIV uses receptors and co-receptors like CCR5, CXCR4, and CD4 to enter the host cell. Some viruses are less particular; for example, cytomegalovirus (CMV) targets host glycoproteins and integrins located on many host cells. Here's a few common viruses we'll discuss in the future and the receptors they bind to:

Table 1

Viru[illegible]	[illegible]ptor(s)
CMV	Integrins (heparan sulfate)
EBV	CD21, found on B cells
HIV	CD4, CXCR4, CCR5
Parvovirus B19	P antigen on red blood cells
Rabies	Nicotinic acetylcholine receptors (nAChRs)
Rhinovirus	ICAM-1
SARS-CoV2	ACE2
Influenza virus	sialic acid receptors

Viral attachment to cell surfaces allows the virus to gain entry through two general mechanisms:

- **Receptor-mediated endocytosis:** the virus signals the cell to effectively engulf (or endocytose) the virus.
- **Fusion of the viral envelope:** the viral envelope fuses with the cell membrane, releasing the capsid into the cell.

CLINICAL CORRELATION

A number of individuals, primarily of Northern European descent, were found to be highly resistant to HIV infections. These individuals have a mutated CCR5 receptor that bars entry to HIV. Some medications used to treat HIV, such as Maraviroc, block CCR5 and inhibit viral entry.

CLINICAL CORRELATION

Ethanol-based hand sanitizers inactivate **enveloped viruses**—like SARS-CoV-2, herpesviruses, influenza, RSV, and parainfluenza—by dissolving their lipid envelope and preventing them from infecting cells.

However, not all viruses are enveloped. **Non-enveloped viruses**, such as noroviruses, remain infectious even after exposure to ethanol-based hand sanitizers. In these cases, washing with soap and water is the most effective way to remove them from your hands.

Where do DNA and RNA Viruses Replicate? (LO3)

There is great variation in where viruses replicate after entering their target cells. We can group the intracellular site of replication based on the same grouping we used previously.

DNA Viruses. With some exceptions, such as poxviruses, DNA viruses replicate in the cell's nucleus. Once they've injected their DNA into the host cell, that DNA is able to enter the nucleus and access the nucleic acid synthesis machinery it needs to produce mRNA and proteins to make more virions.

RNA Viruses. Most RNA viruses primarily undergo replication in the cytoplasm, using their own machinery in combination with host cell ribosomes. Each subgroup follows a different general mechanism:

- **dsRNA viruses:** These viruses introduce multiple RNA molecules coding for multiple prot[illegible]
- **(+)ssRNA viruses:** The genome is directly translated by host ribosomes into a single large protein that can then be processed into smaller proteins.
- **(–)ssRNA viruses:** Most must introduce an enzyme called RNA replicase (also known as RNA-dependent RNA polymerase), which forms (+)ssRNA, which can function like the RNA of (+)ssRNA.

This can be remembered by where the nucleic acid type exists in a host cell. DNA is normally kept in the nucleus, while mRNA is sent out of the nucleus for translation. The same general rule applies for viral replication, with some exceptions. For example, poxviruses, which are dsDNA viruses, replicate in the cytoplasm, while influenza and retroviruses replicate in the nucleus.

Where do DNA and RNA viruses generally replicate?

Retroviruses. Retroviruses function very differently from most RNA viruses. They use the RT enzy[illegible] then integrate into the host genome using an enzyme called integrase. They can then use host machinery to replicate themselves based on that integrated DNA.

CLINICAL CORRELATION

Because (+)ssRNA is effectively equivalent to mRNA, purified (+)ssRNA virus RNA is actually contagious all on its own! If it is able to enter a cell, this mRNA can begin to undergo replication and produce more virus.

How is (-)ssRNA translated into a protein?

How Do Viruses Get Out of the Cell? (LO2,3)

Once new virions hav[illegible] their way out of the cell. This occurs through two primary methods:

- Cell lysis
- Viral budding

Remember, each cell can only hold so much volume. Unchecked viral replication in an infected cell can produce so many virions that they actually cause the cell to burst, or lyse. This process releases a massive quantity of virions into the host's system, either locally or systemically. These released virions can each go on to infect—and eventually lyse—their own cells. This type of infection is called a **lytic infection**.

Other viruses, especially those using host membranes as their envelopes, are released through viral budding. Newly assembled virions move to the cell membrane and form an envelope from host cell membrane. The coating of host membranes gives these viruses a way to evade host defenses. Influenza and respiratory syncytial virus (RSV) are examples of viruses that use host cell membranes in the creation of their envelopes.

What Does a Viral Infection Look Like? (LO3)

So how does a virus go from infecting a single cell to infecting enough cells to affect a host made out of trillions of cells? To understand that, we need to understand the stages of viral infection.

Types of Viral Infection

We've discussed the stages of a single cell's infection with a virus. The last stage, release, can go a few different ways, presenting with variable types of infections.

Persistent Infections. Some viruses can stay within the host cell and continue to replicate over many months or years. This can occur via integration into the host genome, evasion of host defenses, or viral mutations. These infections can continue for very long periods (months to years), resulting in chronic infections. HIV causes persistent infections. HIV incorporates into the host genome, becoming a permanent part of that cell.

Latent Infections. Latent infection occurs when an infection lacks a trigger for replication and the virus lays dormant inside the cell. No virus is produced during this time. When triggered, such as when an infected host undergoes immunosuppressive therapy for an organ transplant, these otherwise harmless latent diseases can quickly spread out of control. This process is called reactivation and occurs with herpesviruses such as herpes simplex virus-1 (HSV-1), which usually presents as cold sores.

In addition to immunosuppression, aging, stress, pregnancy, certain medications, malnutrition, and other infections can cause latent infections to activate.

What is the difference between a persistent and a latent infection?

How are Viral Infections Diagnosed? (LO4)

So if viruses are too small to see even with stains, how can we diagnose them? In addition to recognizing common clinical presentations, there are several general assays used to confirm a diagnose. While each virus may have a preferred method of detection, generally these are the way a diagnosis is confirmed:

- **Nucleic acid application test (NAAT):** NAAT like polymerase chain reaction (PCR) and reverse transcriptase polymerase chain reaction (RT-PCR) can be used to rapidly test patient samples for multiple pathogens at once, including viruses. PCR amplifies pathogen DNA, while RT-PCR makes complementary DNA (cDNA) copies from pathogen RNA, using reverse transcriptase, and then amplifies the cDNA.
- **Serology:** Production of antibodies against viral antigens can indicate that the host is or was infected with a specific pathogen.
- **Antigen-capture assay:** Using antibodies with a tag, pathogen antigens can be observed directly in patient samples.
- **Histology:** Some infections produce unique characteristics in cells and/or tissues, called cytopathic effects. These changes can help narrow down the potential pathogen and will be pointed out as we cover specific pathogens.

While useful, serology has its limitations when diagnosing infections. For starters, early in the [illegible]ot have risen enough for detection. So, a patient could have symptoms, but their antibody titers could be so low that they are below the limit of detection. Thus, early in an infection, a patient could be negative for antibodies but still have an infection.

Recall that IgM is the first antibody class to rise after an infection and that in a few weeks, IgG is the predominant antibody found in serum. IgG can persist in serum for years. The presence of IgM and/or IgG to a pathogen helps to determine current infection and a past infection.

CLINICAL CORRELATION

Many patients who have HIV do not know exactly when they were infected. So, one of the most common screening assays for HIV is an antigen/antibody combination assay.

This assay detects HIV proteins (p24) and/or host antibodies in the blood against HIV (either IgM or IgG). Early in the infection, HIV proteins, like p24, rise in the blood and can be detected by this assay. Later in the infection, the host will produce antibodies against HIV and these antibodies can be detected as well. So regardless of when the patient was infected, this assay has the best chance to successful detecting the virus.

RT-PCR and antigen-capture assays are more sensitive than serology early in infection. Let's use COVID-19 as an example to illustrate when these assays could be useful. Figure 3 shows the infectious cycle of COVID-19 with the viral load (number of virio[illegible]tibody response.

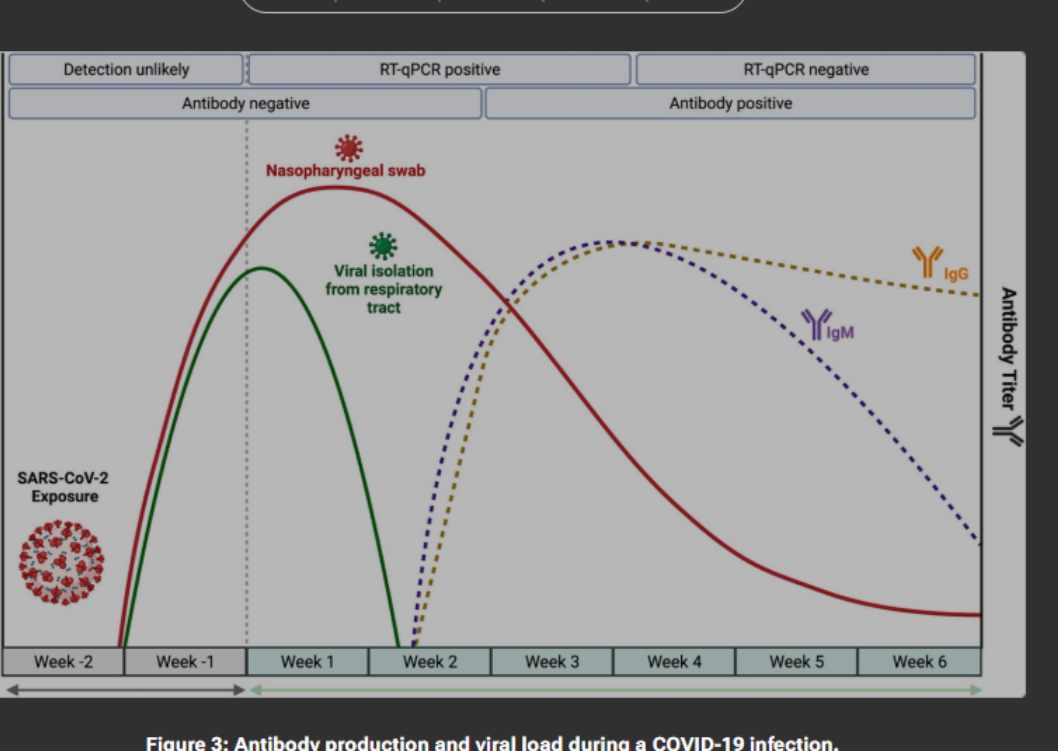


Figure 3: Antibody production and viral load during a COVID-19 infection.

Credit: *ACS Infect. Dis.* 2022, 8, 9, 1758-1814

Note that in Figure 3, the patient can have symptoms for roughly 1-2 weeks before IgM and IgG begin to rise. Thus, serology would not be appropriate during the early phases of the infection. RT-PCR of a patient sample or an antigen-capture assay of a nasal swab would be the best way to confirm a diagnosis of COVID-19 early in the infection.

Note that serology can be very useful in diagnosing persistent or latent infections. For example, HIV and HSV infections can persist for life with undetectable viral loads. As we cover these specific viruses, we will cover if and when these assays ar[illegible]

CASE CONNECTION

BACK TO INTRODUCTION ↑

Our patient has had symptoms for approximately 4 days; it is unlikely that serology would be useful. Recall that it takes weeks for antibody titers to rise enough to confirm an infection.

With infections that are recent, NAAT like RT-PCR or an antigen-capture assay would be the most appropriate test if available. A nasal swab is taken, and the sample is run on a COVID-19 antigen-capture assay. The results are positive. Since the patient is young and has no significant risk factors for severe COVID-19, supportive therapy for symptoms is provided.

Summary

- Viruses are nonliving particles that contain either DNA or RNA and use living cells to reproduce.
- (+)ssRNA viruses can be transcribed using host ribosomes, (-)ssRNA viruses use a viral RNA polymerase to create (+)ssRNA from the (-)ssRNA. Afterward the (+)ssRNA is used to make viral proteins using host ribosomes.
- Retroviruses, like HIV, encode their genetic material as RNA but use reverse transcriptase to convert their RNA into DNA.
- Viral infection involves six steps: attachment, penetration, uncoating, replication, assembly, and release.
- The replication site of viruses varies, with most DNA viruses replicating in the cell's nucleus[illegible]sm.
- Viruses enter cells by binding receptors present on the surface of cells. This determines the trophism of a virus or which cells and tissue they infect.
- Diagnosis of viral infections can be done through methods such as nucleic acid application test (NAAT), serology, antigen-capture assays, and histology.
- Serology is of limited use early in an infection since it takes weeks for antibody titers to rise to a detectable level. Serology is more useful for prolonged or chronic infections.
- Early in a viral infection, NAAT or antigen-capture assays are more consistent if available.
- Some viruses have distinct histological characteristics that can help distinguish them from other infections that may clinically present in a similar manner.

Review Questions

1. A 57-year-old female with metastatic pulmonary adenocarcinoma is enrolled in a study on viral prophylaxis with a new antiviral medication. This medication is designed to target the viral RNA replicase enzyme inside cells, inhibiting viral replication. What type of virus would this antiviral best target?

A. ○ dsDNA virus
B. ○ dsDNA-RT virus
C. ○ ssDNA virus
D. ● (–)ssRNA virus

Hide Explanation

The correct answer is (–)ssRNA virus (D). In this question, a medication is targeting a type of enzyme specifically used by (–)ssRNA viruses to produce (+)ssRNA, which can be used to code for viral components as if it were mammalian mRNA. dsDNA viruses (A) do not use this enzyme, instead using host transcription machinery to produce mRNA. The same is true for dsDNA-RT viruses (B), though these viruses do carry their own polymerase to produce RNA intermediates. ssDNA viruses (C) use a similar replication process as dsDNA viruses, without needing RNA replicase.

2. Leronlimab is a monoclonal antibody that binds to the CCR5 receptor on human T lymphocytes. It is currently in clinical trials to treat HIV. Which step of viral replication does this antibody most likely inhibit?

A. ● Attachment
B. ○ Uncoating
C. ○ Replication
D. ○ Assembly

Hide Explanation

A) Since the antibody binds a host receptor that is recognized by HIV, it blocks the attachment step of viral replication. B), C), and D) occur after attachment, and would not be inhibited by a treatment that binds CCR5. However, other drugs have been developed to inhibit multiple stages of HIV's replication cycle, not just attachment. For example, multiple reverse transcriptase inhibitors are available to impede HIV replication.

3. A 45-year-old male presents with fever, cough, and shortness of breath. On examination, he has a temperature of 102°F (38.9°C), tachycardia, and bilateral crackles on lung auscultation. Chest X-ray reveals bilateral infiltrates. COVID-19 is suspected and has been prevalent in the community for months. The patient has only had symptoms for 4 days. Which test would be the best to confirm the diagnosis?

A. ○ Cell culture
B. ○ Staining and microscopy
C. ○ Serum IgM levels
D. ○ Serum IgG levels
E. ● Antigen-capture assay

Hide Explanation

E) Due to the recent onset of the patient's symptoms, it is likely that viral load is high and detecting either viral antigens (antigen-capture assay) or nucleic acid (RT-PCR) would be the best way to confirm a diagnosis. A) Cell culture is very time consuming and difficult for viruses. It is not standard practice. B) Viruses are very small and stain poorly, thus staining/microscopy is not an optimal choice. C) and D) are wrong. Given the short duration the patient has had symptoms, it is unlikely that IgM or IgG titers are high enough to detect.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe how common structures and genetic information are used to categorizes viruses into families.~~
02 ☑ ~~Explain the key steps of a viral infection.~~
03 ☑ ~~Compare and contrast the replication cycle of DNA, (–)ssRNA, and (+)ssRNA viruses.~~
04 ☑ ~~Compare and contrast the common lab assays used to identify viral infections and their limitations.~~

stain poorly, thus staining/microscopy is not an optimal choice. C) and D) are wrong. Given the short duration the patient has had symptoms, it is unlikely that IgM or IgG titers are high enough to detect.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe how common structures and genetic information are used to categorizes viruses into families.~~
02 ☑ ~~Explain the key steps of a viral infection.~~
03 ☑ ~~Compare and contrast the replication cycle of DNA, (–)ssRNA, and (+)ssRNA viruses.~~
04 ☑ ~~Compare and contrast the common lab assays used to identify viral infections and their limitations.~~

Objective Confidence: 4/4